Computer Workshop Notes

Prepared by Arpan Adhikari

BE First Year | Pokhara University | Faculty of Science and Technology

Simple, easy notes for all 8 units

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Unit 1: Introduction to Desktop / Workstation

A **computer** is a machine that takes data (input), works on it (process), keeps it (storage), and shows the result (output). A **desktop** is a computer that stays on a table. A **workstation** is a more powerful desktop used for heavy work like design or engineering.

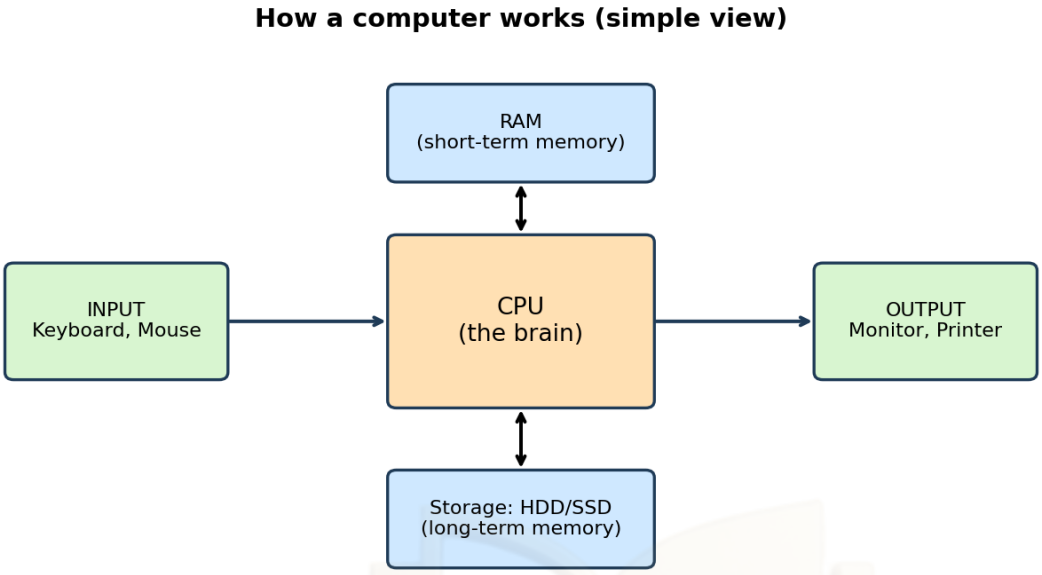


Figure 1.1: Input → Process → Output, with memory and storage

1.1 Main hardware parts

Hardware means the parts of a computer you can touch.

Part	What it does (simple)	Example
Processor (CPU)	The brain. It does all the calculation and follows instructions.	Intel Core i5, AMD Ryzen 5
Motherboard	The main board. All parts connect to it and talk through it.	ASUS, Gigabyte, MSI boards
Memory (RAM)	Short-term memory. Holds work that is open now. Empties when power goes off.	8 GB DDR4, 16 GB DDR5
Storage	Long-term memory. Keeps files even when power is off.	HDD, SSD, pen drive
Input devices	Give data to the computer.	Keyboard, mouse, scanner, mic
Output devices	Show or give results from the computer.	Monitor, printer, speaker

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Main parts on a motherboard

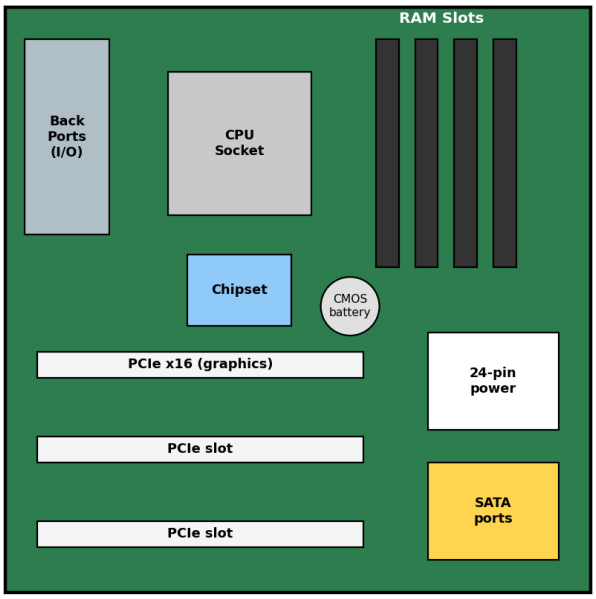


Figure 1.2: Simple layout of a motherboard

Storage devices

- **HDD (Hard Disk Drive):** uses a spinning metal disk. Cheap and big, but slow.
- **SSD (Solid State Drive):** no moving parts, uses chips. Very fast, more costly.
- **NVMe SSD:** a small stick-like SSD that plugs directly into the motherboard (M.2 slot). Fastest.
- **Optical drive (CD/DVD):** reads discs using a laser. Now rarely used.
- **Pen drive / external disk:** portable storage using USB.

1.2 Power supply (SMPS), cables and ports

SMPS means *Switch Mode Power Supply*. It takes AC electricity from the wall (220 V in Nepal) and changes it into low DC voltages that the computer needs: **+12 V, +5 V and +3.3 V**.

Cable / Connector	Goes to
24-pin ATX	Motherboard (main power)
4/8-pin EPS (CPU)	Near the CPU, gives power to the processor
SATA power	Hard disk, SSD, DVD drive
6/8-pin PCIe	Graphics card
Molex (4-pin)	Old drives, fans
SATA data cable	Connects disk to motherboard (data, not power)

Ports are the holes/sockets where we plug devices:

- **USB** – pen drive, mouse, keyboard, phone. (USB-A is the flat one, USB-C is small and oval.)
- **HDMI / DisplayPort / VGA** – to connect a monitor. VGA is old (blue, 15 pins).
- **RJ-45 (LAN)** – network cable for internet.
- **Audio jacks** – green = speaker/headphone, pink = mic, blue = line-in.
- **PS/2** – old round port for keyboard (purple) and mouse (green).

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Remember: CPU = brain, RAM = table where you work, Storage = cupboard where you keep things, SMPS = heart that gives power.

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Unit 2: Assembling a Desktop Computer

Assembling means putting all parts together to make a working computer. **Disassembling** means taking them apart.

2.1 Safety precautions (very important)

- Always **switch off and unplug** the power cable before opening the case.
- Wear an **anti-static wrist strap**, or touch the metal case first. Static electricity from your body can kill chips.
- Hold cards and RAM from the **edges**. Do not touch gold pins.
- Work on a clean, dry, non-metal table. Keep water and food away.
- Never force a part. If it does not fit, check the direction (notch/key).
- Keep screws in a small box so they do not get lost.
- Do not open the SMPS box. It stores high voltage even after unplugging.

2.2 Step-by-step assembling

1. **Step 1: Prepare the case** – Open the side panel. Fit the **standoffs** (small brass screws) and the I/O shield at the back.
2. **Step 2: Install the CPU** – Lift the socket lever. Match the small **golden triangle** on the CPU with the mark on the socket. Place gently (no pressing), close the lever.
3. **Step 3: Apply thermal paste and cooler** – Put a pea-size dot of thermal paste on the CPU. Fix the cooler/fan on top. Connect its cable to **CPU_FAN** header.
4. **Step 4: Install RAM** – Open the clips. Match the notch in the RAM with the slot. Press down firmly on both ends till the clips lock with a 'click'. For 2 sticks, use slots 2 and 4 (check manual).
5. **Step 5: Fix motherboard in case** – Place it on the standoffs, line up with the I/O shield, and screw it.
6. **Step 6: Install storage** – Fix HDD/SSD in the drive bay. Connect **SATA data** cable to motherboard and **SATA power** from SMPS. For M.2 SSD, slide into slot and screw.
7. **Step 7: Install CD/DVD drive (if any)** – Slide into the 5.25" front bay and connect SATA data and power.
8. **Step 8: Install graphics / PCIe card** – Remove the back slot cover. Push the card into the PCIe x16 slot until it clicks. Screw it.
9. **Step 9: Install SMPS** – Screw it at the top or bottom back of the case.
10. **Step 10: Connect power cables** – 24-pin to motherboard, 4/8-pin to CPU area, PCIe power to graphics card.
11. **Step 11: Connect front panel** – Power switch, reset switch, power LED, HDD LED, front USB and audio. Follow the motherboard manual (pins marked PWR_SW, RESET, etc.).
12. **Step 12: Check and test** – Check all cables. Close the case. Connect monitor, keyboard, mouse and power. Switch ON. If you see the BIOS screen, it is working.

Setting of parts (short notes)

- **Motherboard:** Choose one that supports your CPU socket (e.g. LGA 1700 for Intel, AM5 for AMD).
- **Memory:** RAM type must match the board (DDR4 or DDR5). They are not interchangeable.

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- **Hard disk:** In old IDE drives, a **jumper** was set as Master or Slave. SATA drives do not need this.
- **Processor:** Never run a CPU without a cooler. It will overheat in seconds.
- **SMPS:** Choose enough watts (e.g. 450–650 W for normal PC). Some have a 110/230 V switch; in Nepal keep it on **230 V**.

Beep codes (if the PC does not start)

When a computer starts, it tests itself. If something is wrong, it gives **beeps**. Example (varies by company): one short beep = OK; long repeating beeps = RAM problem; one long and three short = graphics problem.

Remember: Most 'dead' new builds are due to: RAM not fully pressed, CPU 8-pin power not connected, or front power switch wire in the wrong pins.

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Unit 3: Assembling a Laptop

A laptop has the same parts as a desktop, but they are small and packed tightly. Many parts (like CPU) are **soldered** on the motherboard, so we cannot change them.

3.1 Steps and precautions

- Shut down, unplug the charger, and **remove the battery** (or disconnect the internal battery cable first).
- Use the right tools: small Phillips screwdriver, **plastic spudger/pry tool** (not metal), tweezers.
- Laptop screws have different lengths. Place them on paper in the same pattern as they came out, or take a photo.
- Ribbon cables (flat cables) are delicate. Lift the small lock tab first, then slide the cable out.
- Do not bend the motherboard. Do not pull the display cable hard.
- Take photos at every step so you can put it back the same way.

General opening order

1. Remove bottom screws (some are under rubber feet or stickers).
2. Gently open the bottom cover with a plastic pry tool, starting from a corner.
3. Disconnect the battery connector.
4. Now you can reach RAM, SSD, Wi-Fi card, fan and heat sink.
5. To reassemble, do the same steps in reverse order.

3.2 Laptop parts compared to desktop

Part	In a laptop
Motherboard	Small, custom shape. Made for one model only.
Processor	Usually soldered (BGA). Cannot be upgraded.
Memory	SO-DIMM (short RAM stick) or soldered. Insert at 45° angle and push down until clips lock.
Hard disk	2.5-inch SATA SSD/HDD or M.2 NVMe SSD.
CD-ROM	Old laptops had a slim DVD drive. New ones do not.
Power (SMPS)	Outside the laptop – the **charger/adapter** (e.g. 19 V DC). Inside there is a battery (Li-ion).
Cooling	Small fan + copper heat pipe. Clean dust regularly.

Remember: Laptop = desktop in a small box. Most common upgrades are only RAM and SSD.

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Unit 4: Laptop / Desktop OS Installation

An **Operating System (OS)** is the main software that runs the computer and lets us use other programs. Examples: **Windows, Linux (Ubuntu), macOS**.

4.1 BIOS, firmware and boot setting

Firmware is small software stored inside a chip on the motherboard. It starts the computer before the OS. There are two types:

	Legacy BIOS	UEFI (new)
Age	Old (since 1980s)	New (used in all modern PCs)
Look	Blue/grey text screen, keyboard only	Graphic screen, mouse can work
Disk type	MBR	GPT
Max disk size	2 TB	Very large (more than 2 TB)
Secure Boot	No	Yes (blocks unsafe boot software)

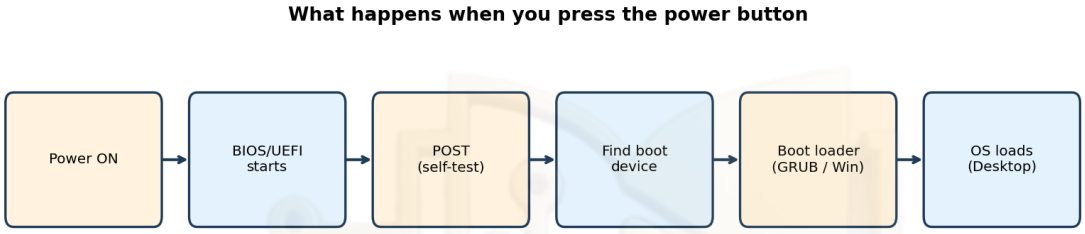


Figure 4.1: Boot process

How to enter BIOS/UEFI setup: Press a key right after power ON – usually **Del, F2, F10, F12 or Esc** (depends on company).

Boot order / Boot priority: The list that tells the computer where to look for an OS first (USB, SSD, DVD, Network). To install an OS from a pen drive, put **USB first**. Many PCs also have a one-time **Boot Menu key** (F12, F11, Esc).

Other BIOS settings: date and time, enable/disable Secure Boot, SATA mode (AHCI), virtualization (VT-x/AMD-V), fan speed. The **CMOS battery** keeps these settings saved.

4.2 Hard disk partitioning

Partitioning means dividing one hard disk into smaller parts. Each part acts like a separate drive (C:, D:, etc.).

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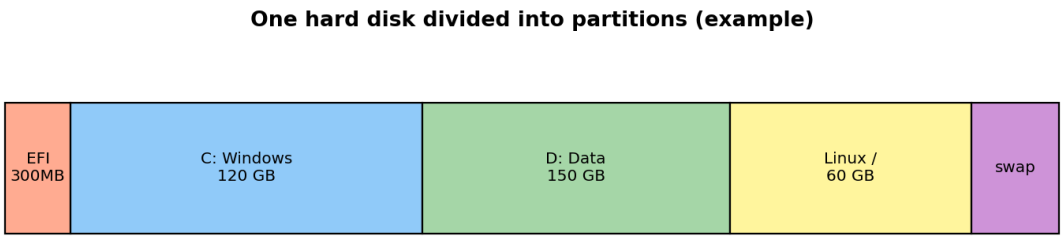


Figure 4.2: Example partitions for dual boot (Windows + Linux)

- **MBR:** old style; max 4 primary partitions and 2 TB.
- **GPT:** new style; up to 128 partitions, works with UEFI.
- **File systems:** Windows uses **NTFS** (FAT32/exFAT for pen drives). Linux uses **ext4**. Linux also uses a **swap** area (extra memory on disk).
- **Formatting** = preparing a partition to store files. It erases data!
- Tools: Windows **Disk Management** (diskmgmt.msc) or **diskpart**; Linux **GParted**, **fdisk**.

4.3 OS installation

First, make a bootable USB: download the OS file (ISO) from the official website, and use **Rufus** or **balenaEtcher** (or Microsoft Media Creation Tool) to write it on an 8 GB+ pen drive.

Windows 10/11 install:

1. Insert USB, restart, choose USB from Boot Menu.
2. Select language, keyboard, click **Install now**.
3. Enter product key (or 'I don't have a key').
4. Choose **Custom install**.
5. Select or create a partition, click Next. Files copy and PC restarts few times.
6. Set region, keyboard, user account and password. Done.

Linux (Ubuntu) install:

1. Boot from USB, choose **Try or Install Ubuntu**.
2. Select language, keyboard, Wi-Fi.
3. Choose Normal installation, tick 'install third-party software'.
4. Choose **Erase disk** (whole disk) or **Something else** (manual partitions for dual boot).
5. Select time zone (Kathmandu), create user name and password.
6. Wait, then restart and remove the USB.

Remember: For dual boot, install **Windows first, then Linux**. Linux's boot loader (GRUB) will show a menu to choose either OS.

4.4 User account management

A **user account** lets each person have their own files and settings with a password.

- **Administrator:** full power – can install software and change settings.
- **Standard user:** can use programs but cannot change important system settings.
- **Guest:** temporary use, very limited.

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Task	Windows	Linux (terminal)
Add user	Settings → Accounts → Other users, or **net user ram 1234 /add**	**sudo adduser ram**
Change password	**net user ram newpass**	**passwd ram**
Make admin	**net localgroup administrators ram /add**	**sudo usermod -aG sudo ram**
Delete user	**net user ram /delete**	**sudo deluser ram**

4.5 Printer and scanner installation

1. Connect the printer by USB, or to the same Wi-Fi/LAN network.
2. Switch it ON. Windows/Linux often find it automatically (plug and play).
3. If not, download the **driver** from the printer company's website (HP, Canon, Epson) and install it.
4. Windows: Settings → Bluetooth & devices → Printers & scanners → Add device. Linux: Settings → Printers → Add.
5. Print a **test page** to check.
6. For a scanner, install its driver/software, then use **Windows Scan** app or Linux **Document Scanner**.

A **driver** is a small program that tells the OS how to talk to a hardware device.

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Unit 5: Networking and Internet Setup

5.1 What is a computer network?

A **computer network** is two or more computers connected together to share data, internet, printers and files.

Type	Area covered	Example
PAN	Very small (around 10 m)	Phone to Bluetooth earbuds
LAN	A room, building, college	Computer lab network
MAN	A city	City cable TV / ISP network in Pokhara
WAN	Countries / whole world	The Internet

Network topologies

Topology means the shape or layout of how computers are connected.

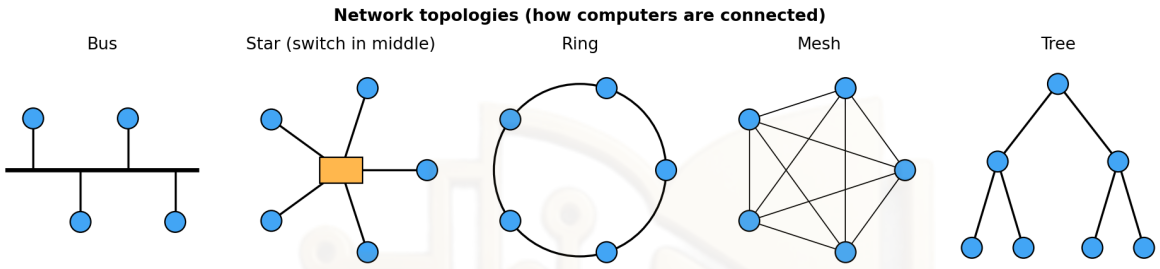


Figure 5.1: Common network topologies

Topology	Good	Bad
Bus	Cheap, less cable	If main cable breaks, all stop
Star	Easy to add PCs; one PC fails, others work	If switch fails, all stop
Ring	Data flows in order	One break can stop the ring
Mesh	Very reliable, many paths	Costly, too many cables
Tree	Easy to expand large networks	Top link failure affects branch

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Most offices and homes today use the **Star** topology.

Networking media

- **Wired (guided):** twisted pair cable (LAN cable), coaxial cable (cable TV), optical fibre (light signals – very fast, used by ISPs like FTTH).
- **Wireless (unguided):** Wi-Fi (radio waves), Bluetooth, infrared, microwave, satellite.

5.2 Network cabling, cable types and connectors

Cable	Speed (approx.)	Notes
Cat5e	1 Gbps	Most common in labs
Cat6	1 Gbps (10 Gbps short)	Better, less noise
Cat6a / Cat7	10 Gbps	For fast/data-centre use

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Cable	Speed (approx.)	Notes
Coaxial	–	Copper core, used for TV/old networks. BNC / F connector
Fibre optic	Very high (10 Gbps+)	Glass/plastic, light. SC / LC connector

- **UTP** = Unshielded Twisted Pair (normal). **STP** = Shielded (has foil to stop noise).
- Wires are twisted in pairs to reduce **noise (interference)**.
- Max length of one LAN cable = **100 metres**.

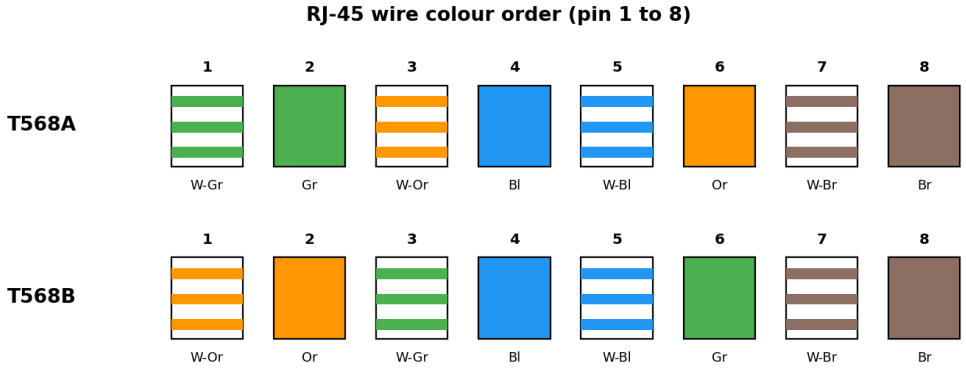


Figure 5.2: T568A and T568B colour codes for the RJ-45 connector

Straight-through cable: both ends same (B–B). Used for **different** devices: PC ↔ switch, router ↔ switch.

Crossover cable: one end A, other end B. Used for **same** devices: PC ↔ PC, switch ↔ switch. (Modern devices with Auto-MDIX work with either.)

How to crimp a LAN cable:

1. Strip about 2–3 cm of outer jacket with a stripper.
2. Untwist the pairs and arrange colours in T568B order.
3. Cut the ends straight, leave about 1.2 cm.
4. Push wires fully into the RJ-45 connector (clip facing down).
5. Press with the **crimping tool**.
6. Test with a **LAN cable tester** – all 8 lights should blink in order.

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Network devices

- **NIC (Network card):** lets the computer join a network. Every NIC has a unique **MAC address**.
- **Hub:** old; sends data to all computers (wasteful).
- **Switch:** smart; sends data only to the right computer (uses MAC address).
- **Router:** connects different networks, e.g. your LAN to the Internet (uses IP address).
- **Access Point (AP):** gives Wi-Fi to devices.
- **Modem / ONT:** changes ISP signal (phone line or fibre) into usable internet.

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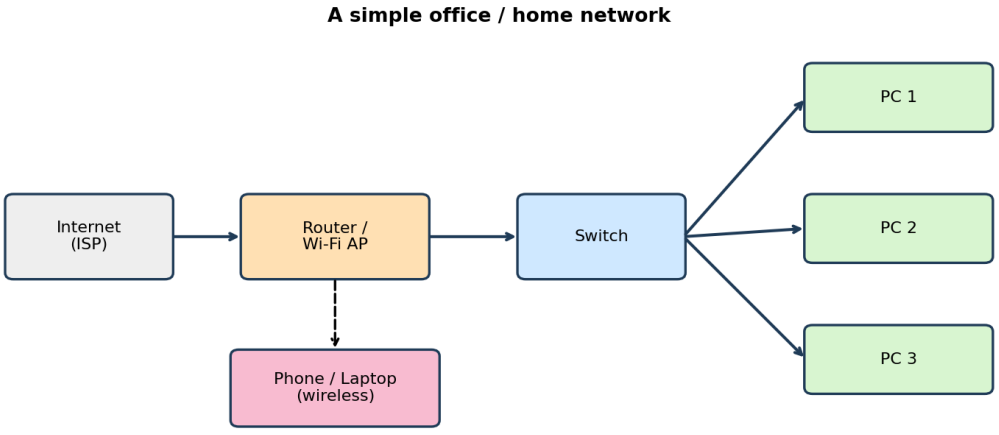


Figure 5.3: A simple network with wired and wireless devices

5.3 Basic wireless AP / router configuration

1. Connect PC to the router using LAN cable (or its default Wi-Fi).
2. Open a browser and type the router address, e.g. **192.168.0.1** or **192.168.1.1** (written on the router sticker).
3. Log in with default user/password (often admin/admin). **Change this password first.**
4. WAN / Internet setting: choose type given by ISP (PPPoE with username/password, or Dynamic IP).
5. Wireless setting: set **SSID** (Wi-Fi name) and choose band (2.4 GHz = longer range, 5 GHz = faster).
6. Security: choose **WPA2 or WPA3**, set a strong Wi-Fi password. Never use 'Open' or old 'WEP'.
7. DHCP: keep ON so devices get IP addresses automatically.
8. Save, reboot, and connect your phone/laptop to test.

Remember: Switch = connects devices inside a network. Router = connects your network to the Internet.

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Unit 6: Software Installation

Software is a set of programs that tells the computer what to do. You cannot touch software.

Type	Meaning	Examples
System software	Runs the computer	Windows, Linux, drivers, BIOS
Application software	Does a job for the user	MS Word, Chrome, VLC, AutoCAD, VS Code
Utility software	Keeps the computer healthy and tidy	Antivirus, WinRAR/7-Zip, Disk Cleanup, backup tools

6.1 Installing application and utility software

On Windows:

- Download the setup file (.exe or .msi) from the **official website** only.
- Double-click → accept licence → choose folder → Install → Finish.
- Or use Microsoft Store, or command: **winget install VideoLAN.VLC**
- To remove: Settings → Apps → Installed apps → Uninstall.

On Linux (Ubuntu):

- Use the App Center, or terminal:
- **sudo apt update** – refresh the software list
- **sudo apt install vlc** – install
- **sudo apt remove vlc** – uninstall
- **sudo dpkg -i file.deb** – install a downloaded .deb file

Before installing, check the **system requirements** (OS version, 32/64-bit, RAM, disk space).

6.2 Updates and firmware patches

An **update / patch** is a small fix from the company. It closes security holes, fixes bugs and adds features.

- **Windows:** Settings → Windows Update → Check for updates.
- **Linux: sudo apt update && sudo apt upgrade**
- **Firmware (BIOS/UEFI) update:** download from the motherboard/laptop company site. Update using BIOS tool (e.g. EZ Flash, Q-Flash) or company app.
- **Drivers:** update from Device Manager or the company website.

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Remember: Never switch off power during a BIOS update. A power cut can make the motherboard dead ('bricked'). Keep laptop on charger.

6.3 Virus scanning

Malware means bad software. Types:

Type	What it does
Virus	Attaches to files and spreads when the file is opened
Worm	Spreads by itself through networks
Trojan	Looks like good software but does harm secretly

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Type	What it does
Ransomware	Locks your files and asks money to unlock
Spyware / Keylogger	Secretly watches you, steals passwords
Adware	Shows unwanted ads

Antivirus finds and removes malware. Examples: Windows Defender (built-in), Kaspersky, ESET, Avast, ClamAV (Linux).

- **Install:** download from the official site → install → let it update its virus database.
- **Update:** keep 'automatic updates' ON; new viruses come every day.
- **Quick scan:** checks common places (fast). **Full scan:** checks everything (slow). **Custom scan:** choose a pen drive or folder.
- Found virus → **Quarantine** (lock it) or **Delete**.
- Linux: **sudo apt install clamav**, then **sudo freshclam** (update) and **clamscan -r /home** (scan).

Stay safe: do not open unknown email attachments, scan pen drives before opening, and avoid cracked software.

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Unit 7: Server Installation and Backup

7.1 What is a server?

A **server** is a computer that gives a service to other computers. The computers that ask for the service are called **clients**. Servers usually run 24 hours, have more RAM and storage, and often use **RAID** (many disks together for safety) and backup power (UPS).

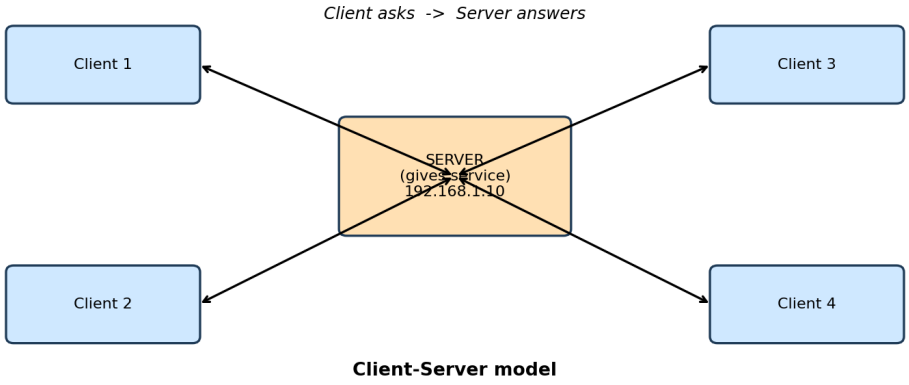


Figure 7.1: Client-server model

Common server OS: Windows Server, Ubuntu Server, Red Hat / Rocky Linux.

7.2 IP address setting

An **IP address** is the address of a computer on a network, like a house number. Example: **192.168.1.10** (IPv4 – four numbers from 0 to 255).

Term	Simple meaning	Example
IP address	Address of the device	192.168.1.10
Subnet mask	Tells which part is the network	255.255.255.0
Default gateway	The router address to go outside	192.168.1.1
DNS server	Changes names (google.com) into IP addresses	8.8.8.8, 1.1.1.1

- **Static IP:** set by hand, never changes. Servers and printers should use static IP.
- **Dynamic IP (DHCP):** given automatically by the router. Used by normal PCs and phones.
- **Private IP:** used inside LAN (192.168.x.x, 10.x.x.x, 172.16–31.x.x). **Public IP:** used on the Internet.

Setting static IP in Windows: Control Panel → Network and Sharing Center → Change adapter settings → right-click Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4) → 'Use the following IP address' → fill IP, mask, gateway, DNS → OK.

Linux: Settings → Network → IPv4 → Manual. Or on Ubuntu Server, edit the **netplan** file in /etc/netplan/ and run **sudo netplan apply**. Check with **ip a**.

7.3 Types of servers

Server	Job	Common software	Port
Email server	Sends and receives email	Postfix, Microsoft Exchange, hMailServer	SMTP 25/587, IMAP 143/993, POP3 110

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Server	Job	Common software	Port
Data / File server	Stores shared files for everyone	Windows File Sharing (SMB), Samba, NFS, NAS	445 (SMB)
Domain server	Controls user logins and computers in one place	Active Directory (Domain Controller), DNS server	389 (LDAP), 53 (DNS)
FTP server	Upload/download files	FileZilla Server, vsftpd, IIS FTP	21 (FTP), 22 (SFTP)
Web server	Hosts websites	Apache, Nginx, IIS	80, 443

Using FTP / SCP client tools

- **FTP:** File Transfer Protocol – moves files, but sends password in plain text (not safe).
- **SFTP / SCP:** safe versions that work over **SSH** (encrypted).
- **FileZilla (client):** enter Host (IP), Username, Password, Port → Quickconnect. Left side = your PC, right side = server. Drag files to upload/download.
- **WinSCP:** Windows tool for SFTP/SCP, works the same way.
- **Command line: scp report.pdf user@192.168.1.10:/home/user/** (upload); **scp user@192.168.1.10:/home/user/data.zip .** (download).

Data backup and restore

Backup = making a copy of data so we do not lose it. **Restore** = bringing the data back from the copy.

Backup type	What is copied	Good / Bad
Full	Everything, every time	Easy restore, but slow and big
Incremental	Only changes since the **last backup**	Fast, small; restore needs all pieces
Differential	Only changes since the **last full backup**	Middle; restore needs full + last differential

3-2-1 rule: keep **3** copies of data, on **2** different media, with **1** copy off-site (another place or cloud).

- **Windows tools:** File History, Backup and Restore, System Restore point, OneDrive.
- **Linux tools: rsync**, **tar**, Timeshift, Deja Dup.
- Example: **rsync -av /home/ram/ /mnt/backup/ram/** copies Ram's files to the backup disk.
- **Restore:** open the backup tool → choose date/version → Restore. Always **test** that restore works.

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Unit 8: Hardware and Software Troubleshooting

Troubleshooting means finding what is wrong and fixing it, step by step.

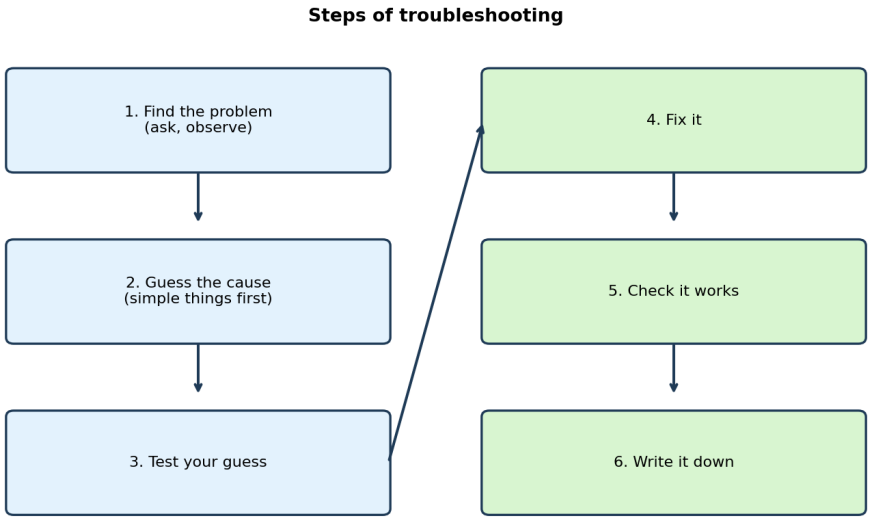


Figure 8.1: Troubleshooting steps

8.1 Basic repair and maintenance

Problem	Possible cause	What to do
PC does not turn on	No power, bad SMPS, loose cable, bad switch	Check wall socket and cable, test SMPS, check front panel wires
Turns on but no display	RAM loose/dirty, graphics card, monitor cable	Remove RAM, clean gold pins with an eraser, reinsert; check monitor cable/input
Beeps at start	Hardware fault (RAM/graphics)	Check beep code in the motherboard manual
Very slow	Low RAM, full disk, virus, too many startup apps, HDD	Disk cleanup, antivirus scan, disable startup apps, upgrade to SSD/more RAM
Restarts or shuts down suddenly	Overheating, bad power	Clean dust, change thermal paste, check fan, use UPS
Wrong time every start	CMOS battery dead	Replace CR2032 battery
No internet	Cable, Wi-Fi, IP setting, router	Check cable/lights, run ping and ipconfig, restart router
Blue screen (BSOD)	Driver problem, bad RAM, bad disk	Note the error code, update/roll back driver, test RAM and disk

Regular maintenance:

- Clean dust from fans and vents with a blower every few months.
- Change thermal paste every 2–3 years.
- Keep OS and antivirus updated. Take backups.
- Use a **UPS** or stabilizer to protect from power cuts and voltage changes.
- Do not block laptop air vents (don't use on bed/pillow).

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8.2 Replacement of passive components

Passive components do not need power to control other signals; they only resist, store or pass energy. Main ones: **resistor, capacitor, inductor**, and also fuse.

Component	Job	Unit	How it fails
Resistor	Controls/limits current	Ohm (Ω)	Burns (black mark), open circuit
Capacitor	Stores charge, smooths power	Farad (F), usually µF	**Bulges** on top or leaks liquid
Inductor (coil)	Stores energy in magnetic field, filters	Henry (H)	Burns or cracks
Fuse	Breaks the circuit if current is too high	Ampere (A)	Wire inside melts

Tools for component work

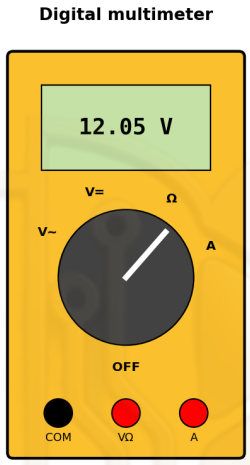


Figure 8.2: A digital multimeter

- **Multimeter:** measures **Voltage (V)**, **Current (A)** and **Resistance (Ω)**. Black probe in COM, red probe in VΩ. **Continuity test (beep):** checks if a wire or fuse is OK – beep means connected.
- **Soldering iron:** a hot pen (about 300–350 °C) that melts **solder** (metal wire) to join components to the board.
- **Desoldering pump / wick:** removes old solder.
- **Hot air gun:** blows very hot air to remove or fit tiny surface-mount (SMD) parts and chips.
- **Flux:** helps solder flow and stick nicely.

Steps to replace a bad capacitor:

1. Unplug power. Wait and discharge big capacitors (they can hold charge).
2. Note the value (e.g. 1000 µF, 16 V) and the **polarity** (– side stripe).
3. Heat the pins with the soldering iron and remove old solder with a pump/wick.
4. Pull out the old capacitor.
5. Insert a new one with the **same capacitance** and **same or higher voltage**, with correct polarity.
6. Solder the pins, cut extra legs, check with multimeter.

Remember: Soldering safety: work in open air (fumes), keep iron on its stand, never touch the tip, wear eye protection.

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8.3 Basic software troubleshooting commands

Windows command	Use	Linux command
ipconfig (/all, /release, /renew)	See IP address and network info	**ip a**, **ifconfig**
ping google.com	Check if a device/site is reachable	**ping google.com**
tracert google.com	See the path data takes	**traceroute google.com**
nslookup google.com	Check DNS (name to IP)	**nslookup**, **dig**
netstat -an	See network connections and ports	**ss -tuln**
sfc /scannow	Find and repair broken Windows files	**sudo apt --fix-broken install**
chkdsk C: /f	Check and fix disk errors	**sudo fsck /dev/sda1**
DISM /Online /Cleanup-Image /RestoreHealth	Repair Windows image	–
tasklist / **taskkill /IM app.exe /F**	List / stop running programs	**ps aux** / **kill PID**, **top**
systeminfo	Show system details	**uname -a**, **lscpu**, **free -h**, **df -h**
ipconfig /flushdns	Clear DNS memory	**resolvectl flush-caches**

8.4 Troubleshooting tools and utilities

- **Task Manager** (Ctrl + Shift + Esc): see CPU, RAM, disk use; end stuck programs; turn off startup apps.
- **Device Manager:** see all hardware. A yellow mark means driver problem – update or reinstall driver.
- **Event Viewer:** shows logs of errors and warnings.
- **Safe Mode:** starts Windows with only basic drivers – good for removing bad drivers or viruses.
- **System Restore:** takes the PC back to an earlier working time.
- **Windows Memory Diagnostic / MemTest86:** checks RAM for errors.
- **CrystalDiskInfo:** checks hard disk health (S.M.A.R.T.).
- **HWMonitor / CPU-Z:** show temperature, voltages and hardware details.
- **Linux: top/htop** (running programs), **dmesg** and **journalctl** (logs), **lsblk** (disks), **lsusb/lspci** (devices).
- **Bootable rescue USB** (e.g. Hiren's BootCD PE, Ubuntu live USB): starts a PC even when its OS is broken, to rescue data.

Remember: Always start with simple checks: Is it plugged in? Is it switched on? Did you restart? Most problems are solved here.

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